

Deep Learning Model for Automated Detection and Classification of Central Canal and Neural Foraminal Stenosis at Cervical Spine MRI

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Abstract

Objectives: Developing a Deep learning (DL) model to automatically detect and classify cervical canal and neural foraminal stenosis on cervical spine MRI can improve the accuracy and efficiency of its diagnostic.

Methods: A method for cervical spinal stenosis was proposed based on the DL model, consisting of region of interest (ROI) detection and cascade prediction. First, three part-specific convolutional neural networks were used to detect the ROIs in different parts of cervical MRI images. Then, the cascade prediction of stenosis categories was performed to obtain the results of stenosis level and position on each slice of the patients. Finally, in the testing, the results were fused to obtain a patient-level diagnostic report. The performance was evaluated with the metrics of accuracy (ACC), area under curve (AUC), sensitivity, specificity, F1 Score, and diagnosis time of the DL model, as well as recall rate for ROI detection localization.

Results: The average recall rate of ROIs localization reached 89.3% (neural foramen), and 99.7% (central canal) under the five-fold cross-validation of our DL model. In dichotomous classification (normal or mild vs moderate or severe), ACC and AUC of the DL model were very close to the level of radiologists, and the F1 score (84.8%) of the DL model was slightly higher than that of radiologists (83.8%) at central canal.

Conclusion: The DL model showed comparable performance with subspecialist radiologists for detection and classification of the central canal and neural foraminal stenosis at cervical spine MRI with significant time-saving.

Introduction

Cervical spinal stenosis (CSS) refers to abnormal stenosis of the cervical canal[1]. CSS may lead to changes in cervical spinal cord function and cervical radiculopathy[2]. CSS can affect two different anatomic sites, the central canal and the neural foraminal canal. First, the cervical spinal canal stenosis (SCS) is defined as an abnormal narrowing of the central canal, which lead to compression of the spinal cord, usually resulting in corresponding clinical symptoms. Secondly, the neural foramen stenosis (NFS) can lead to compression of nerve roots because they exit the spinal cord through the neural foramen. Herniated discs, hypertrophy of facet joints and ligaments, or vertebral instability can cause NFS.

Cervical degenerative changes can lead to significant clinical morbidity resulting in cervical SCS and NFS[3]. Patients with CSS have a variety of clinical manifestations, which often make early diagnosis difficult[4]. However, early treatment is essential to keep track of the course of the disease and to prevent persistent deterioration[5]. In addition, the function of the patient at the time of diagnosis and the underlying etiology determine the prognosis of patient with CSS[6].

Conventional cervical MRI is an important tool in the assessment of CSS for the assessment of CSS. It can accurately assess the degree of SCS and NFS and plays a role in determining appropriate treatment, but describing this information in detail in a report can be repetitive and time consuming. A fast and reliable imaging technique is needed to diagnose and quantify different degrees of CSS spinal cord compression. Deep learning (DL) can improve productivity and consistency of diagnostic reporting. DL has also been successfully applied to perform automatic vertebral segmentation of spinal imaging[7, 8].

Most previously published works on automated analysis of degenerative spine imaging have focused on the lumbar spine. For example, DL in the lumbar spine, MRI has been used for vertebral segmentation and intervertebral disc degeneration grading[9–11], a multitask architecture (denoted as SpineNet) has been developed for automated classification of lumbar central canal stenosis and other spinal conditions. A binary classification (present or absent) was used for central canal stenosis. Lu et al. [12] developed a DL algorithm to grade lumbar spinal stenosis at the central canal and neural foramina with no predefined grading system. Most recently, researchers have made use of DL model for automatically detect and classify central canal, lateral recess, and neural foraminal stenosis of lumbar spine [13, 14]. However, MRI studies on cervical spondylosis are relatively few. Merali et al. developed a convolutional neural network (CNN) model to detect cervical spinal cord compression in patients with degenerative cervical myelopathy on MRI scans. The results showed that the heterogeneous group achieved a higher model performance, with an AUC of 0.94[15]. Ma et al. developed a DL model based on MRI scanning for cervical spinal cord compression in patients with cervical disc degenerative diseases and traumatic spinal cord injuries [16].

Currently, to the best of our knowledge, there is no DL model developed to assess stenosis at two region of interests (ROIs) along the cervical spine. Such a model could provide the reporting radiologist with an accurate and reliable diagnostic tool for CSS. In addition, the performance of traditional deep learning methods is unsatisfactory, and traditional CNN has several defects[17]. To address this, some novel algorithms capable of powerful processing in object detection have been proposed. An example of such an algorithm is Faster R-CNN[18], which offers advantages in terms of accuracy and detection speed. Faster R-CNN may improve the possibility of diagnosing lesions from cervical MRI.

Therefore, we aimed to develop a DL model (Faster-RCNN) to automatically detect and classify central canal and neural foraminal stenosis in the cervical spine using axial T2-weighted MRI imaging in this study.

Materials and Methods

The study design was approved by the appropriate ethics review board, which waived the requirement of informed consent due to the retrospective nature of the study.

Patient Datasets

The patients with imaging-diagnosed degeneration were selected from the radiological reporting system of our hospital’s cervical spine MR image database from January 2016 to December 2018. In total, 796 patients (mean age ± standard deviation, 51 years ± 10.06; men: 495, 63.57%) were evaluated. Adult patients (> 18 years) were included. Patients with instrumentation or patients with other diseases (such as spinal tumors, infection, trauma, or scoliosis) were excluded.

MRI Imaging

MRI cervical spine studies were scanned using different MRI scanners (GE 1.5- and 3.0-T platforms; Siemens 3.0 T platforms; United Imaging 3.0 T platforms) with the same sequences and standard phased-array surface coils. Table 1 provides details on the MRI scanners and sequences.

Table 1
Details on the MRI scanners and sequences

MRI scanners	Sequences	FOV (mm ²)	Slice thickness (mm)	TR(ms)/TE (ms)	Flip angle
GE 1.5 T	T2* MERGE	200×150	4	433.1/5.5	20
GE 3.0 T (HDXT)	T2 FRFSE	256×128	4	552.9/12.0	20
Siemens 3.0 T(Prisma)	T2 FRFSE	190×140	4	400.0/17.0	20
United Imaging 3.0 T					
uMR780	GRE	200×150	4	592.1/13.45	20
uMR880	GRE	200×150	4	608/13.33	20

Data Set Labelling

All data were manually labelled by four junior radiologists and two expert musculoskeletal radiologists with over 20 years of experience (served as the reference standard). Each radiologist was blinded to patient demographics and clinical history. With the use of open-source annotation software, bounding boxes were drawn to segment the ROIs (central canal and neural foramina) at and between each cervical disk level. When drawing each bounding box, the annotating radiologist classified the cervical canal stenosis as one of four categories: normal, mild, moderate, or severe. The severity of foraminal stenosis in the cervical spine can be assessed using a three-tiered grading system: normal, mild, moderate, or severe. These were done using well-established criteria for CSS at the central canal and neural foramin[19–22] (Table 2).

Table 2
Evaluation of the severity of CSS by using three-tiered grading system

Degree of spinal stenosis	Neural foramen	Central canal
Normal	2158	1576
Mild	2350	1263
Moderate	2134	654
Severe	734	195
Total	7376	3688

DL Model Development

A method for cervical spinal stenosis was proposed based on the DL model (Faster-RCNN)[18], consisting of an ROI detection module and cascade classification prediction. First, three types of CNNs were used to detect the ROIs in different parts of cervical MRI images (left neural foramen, right neural foramen, central canal). Then, the image features of ROIs extracted by CNNs are fed to the full connection layer (normal, mild, moderate, and severe) for classification. The loss function includes ROIs detection loss and cross-entropy loss function to evaluate the classification situation. Then, cascade classification is performed to predict the CSS status of each slice of the patient. Finally, in the testing, the results were fused to obtain a patient-level diagnostic report (Fig. 1).

After experimental comparison, the optimizer for the DL models used SGD with an initial learning rate of 0.02, each model was trained for 50 epochs, and the learning rate adjustment strategy selected as StepLR, which was updated every 5 epochs, with a gamma of 0.5.

Statistical Analysis

All statistical analyses were performed using SPSS version 23 (IBM SPSS Statistics, USA). Continuous variables are described using mean $\pm$ standard deviation, while categorical variables are described using frequency and ratio. Cohen's κ test was performed to determine the consistency between DL model results and junior radiologist annotated results, providing Cohen's κ coefficient and its 95% confidence interval. The DL model was developed in Python (version 3.8) PyTorch (version 1.8.1).

The performance was evaluated with metrics of accuracy (ACC), the area under the curve (AUC), sensitivity, specificity, F1 score, and diagnosis time being used for classification, as well as recall rate for ROI detection localization. Intersection over Union (IoU) is used to measure the degree of overlap between the predicted area and the annotated area. IoU > 0.5 is considered a reasonable prediction result in most general cases, and the positioning of ROIs is quite accurate. To make full use of the data and avoid the impact of uneven distribution of data itself, a similar five-fold cross-validation is adopted to test the evaluation metrics which can also verify the robustness of the DL model. All test results were averaged over five random partitions that covered each other. One of them was selected as the test set

for each division, and the other four were selected as the training set. All evaluation metrics are the average results of five different divisions.

Results

ROI Detection

Considering $IoU > 0.5$ as the threshold, the average recall rate of ROIs localization reached 89.3% (neural foramen) and 99.7% (central canal) under the five-fold cross-validation of our DL model (Table 3).

Table 3 ROI Recall rate ($IoU > 0.5$)		
Recall rate ($IoU > 0.5$)	Neural foramen	Central canal
DL model a	90.7%	99.7%
DL model b	87.7%	99.9%
DL model c	90.4%	99.9%
DL model d	90.0%	99.3%
DL model e	87.8%	99.9%
Average	89.3%	99.7%

Classification test results

In dichotomous classification (normal or mild vs. moderate or severe), the average accuracy of the five-fold cross-validation of the DL model was 0.880 and 0.827 in the neural foramen and central canal, respectively, which were close to the radiologists' accuracy of 0.909 and 0.842, with a difference of 0.029 and 0.015 (Table 4).

Table 4
The ACC on dichotomous classification A (normal or mild vs moderate or severe

	Parameter	Neural foramen	Central canal
Split a	DL model a	0.864	0.799
	radiologist	0.899	0.838
Split b	DL model b	0.891	0.839
	radiologist	0.915	0.850
Split c	DL model c	0.874	0.820
	radiologist	0.908	0.818
Split d	DL model d	0.887	0.842
	radiologist	0.906	0.842
Split e	DL model e	0.886	0.836
	radiologist	0.916	0.864
Average	DL model	0.880	0.827
	radiologist	0.909	0.842

For the neural foramen, the AUC for the DL model was 0.890 (95% CI: 0.882, 0.8979), slightly higher than the radiologist's 0.838 (95% CI: 0.829, 0.846) (Fig. 2). The sensitivity of the DL model and the radiologist were 0.705 and 0.709, respectively, which were almost identical. The specificity of the DL model was also as high as 0.927, which was well comparable to the radiologist's performance of 0.966.

For the central canal, the test results differed somewhat from those for the neural foramen (Table 5). The mean sensitivity of the DL model (0.835) was 0.11 higher than that of the radiologists (0.725), and although its specificity was lower (0.819 compared to 0.993), the overall AUC (0.876, 95% CI: 0.865, 0.886) was still comparable to the radiologists' performance (0.859, 95% CI: 0.848, 0.870) comparable or even slightly higher (Fig. 3). For the F1 score, the DL model was 0.848, slightly higher than the radiologists' 0.838.

Table 5
Comparison of Performance Metrics

Metric	Parameter	Neural foramen	Central canal
AUC (95% CI)	DL model	0.890 (0.882, 0.897)	0.876 (0.865, 0.886)
	radiologist	0.838 (0.829, 0.846)	0.859 (0.848, 0.870)
Sensitivity	DL model	0.705	0.835
	radiologist	0.709	0.725
Specificity	DL model	0.927	0.819
	radiologist	0.966	0.993
Precision	DL model	0.758	0.839
	radiologist	0.859	0.993
F1 score	DL model	0.718	0.848
	radiologist	0.777	0.838

Table 6
Cohen's κ test (95% CI)

	Neural foramen	Central canal
DL model	0.644 (0.630, 0.657)	0.649 (0.624, 0.674)
radiologist	0.725 (0.714, 0.736)	0.692 (0.670, 0.714)

For the neural foramen (N) and central canal (C) classification, the DL model agreement (N: Cohen's κ = 0.644; 95% CI: 0.630, 0.657; C: Cohen's κ = 0.649, 95% CI: 0.624, 0.674; $P < 0.001$; respectively) were reduced compared with radiologist (N: Cohen's κ = 0.725; 95% CI: 0.714, 0.736; $P < 0.001$; C: Cohen's κ = 0.692; 95% CI: 0.670, 0.714; $P < 0.001$; respectively).

Discussion

Cervical spine MRI is an essential tool for the evaluation of CSS and can accurately assess of SCS and NFS. For patients with suspected CSS, clinical history and examination are the most important diagnostic tools, but spinal neuroimaging is also very critical to confirm and further refine the diagnosis. The grade of CSS is also important, and the degree of stenosis in each region plays an important role in

determining the appropriate treatment. However, describing this information for doctors to check the MR images slide by slide to classify CSS grades in detail in a report can be repetitive and time consuming. Each doctor may have different opinions for the diagnosis of CSS grades.

DL model can be used to automatically detect and classify SCS and NFS on cervical spine MRI, which can improve the accuracy and efficiency of diagnosis and save valuable time of doctors. In our study, a method for CSS was proposed based on the DL model, consisting of ROI detection and cascade prediction at MRI. In dichotomous classification (normal or mild vs moderate or severe), ACC and AUC of the DL model were very close to the level of radiologists, and the F1 score of the DL model was slightly higher than that of radiologists at central canal. The IoU of ROI detection showed it could accurately locate the central canal and neural foramen. Diagnosing whether a slice's central canal or neural foramen is narrowed by cervical MRI sequences takes an average of 15 seconds for radiologists and 0.098 seconds for our DL model (using Intel(R) Xeon(R) CPU E5-2620 v3 @ 2.40GHz as CPU and GeForce GTX TITAN X as GPU). To the best of our knowledge, no prior study has assessed the automated detection and classification of the central canal and neural foramen stenosis.

According to the grading system reported in the previous research[19–22], all subjects were classified into four grades of SCS. Grade 0 with no visible spinal stenosis, grade1 (mild) with no compression of the spinal cord, grade 2 (moderate) with deformation of the spinal cord, and grade 3 (severe) with compression of the spinal cord, with increased signal intensity near the compression level on T2WI. Grade 3 SCS are associated with adverse clinical outcomes and decreased response to decompression surgery [23–25]. Classification systems for NFS were identified for the cervical spine based on MRI oblique sagittal image. Grade 0 shows no stenosis and no perineural fat obliteration. Grade 1 (mild) depicts below 50% of nerve root circumference perineural fat obliteration. Grade 2 (moderate) depicts above 50% perineural fat obliteration. Grade 3 (severe) shows a morphologically collapsed nerve root. Severe perineural fat obliteration is also combined [21]. The degree of SCS and NFS plays a role in determining appropriate treatment. Takahashi et al.[26] showed that the severity of spinal canal distortion with cord compression was directly proportional to the severity of clinical impairment, and Grade 3 SCS had poorer prognosis. Sun et al.[27] showed that the grade of NFS had an important factor in determining the additional uncinete process resection in anterior cervical discectomy and fusion.

Faster R-CNN has been proposed as a new algorithm with strong processing speeds in object detection since 2015. In addition, numerical experiments have shown that the Faster R-CNN combined with novel CNN models has better recognition performance than several traditional detection methods[28]. Most previously published works on automated analysis of spinal stenosis have focused on the lumbar spine. Only a few studies have applied DL models to cervical spine images. Ma et al. implemented a Faster R-CNN to detect lesions on cervical MRI images. The mean average precisions (mAPs) for Faster R-CNN with ResNet-50 and VGG-16 were 88.6 and 72.3%, and the average time of diagnosis was 0.22 and 0.24 s/image, respectively. Experimental results showed that Faster R-CNN improves the possibility of diagnosing lesions from cervical MRI. Merali et al. developed a DL model to detect cervical spinal cord compression in 289 patients with degenerative cervical myelopathy in MRI scans, and achieved high

performance with an AUC of 0.94, and the sensitivity of 0.88, specificity of 0.89, and F1 score of 0.82 on holdout dataset[15]. Our DL model not only has high accuracy but also has a short average diagnostic time and good diagnostic performance. In addition, the size of the data set we used is larger, and the robustness test of the algorithm shows better generalization performance. In addition, there have been few previous studies evaluating the automatic detection and classification of cervical spinal stenosis, and our DL model can provide more flexible and detailed diagnosis results according to clinical needs.

Some limitations of this study should be acknowledged. First, the data set used in this study was limited to the imaging system of our hospital. Although this method can ensure the uniformity of data, compared with other databases, the amount of data is insufficient and the scalability is poor. Future studies should focus on multicenter data. Second, we only focused on axial T2WI as the target of grading for the evaluation of CSS. But the central canal stenosis grading system we adapted selected sagittal T2WI images and the cervical neural foraminal stenosis grading system selected oblique sagittal images. As we all know, the axial T2WI was the single most important sequence for the evaluation of CSS. In addition, a previous study has found that both axial and oblique sagittal images supported strong interobserver reliability to assess the concordance between MRI grading for cervical neural foraminal stenosis[29]. Therefore, it is feasible to evaluate the stenosis of the spinal canal and foramina with an axial position instead of a sagittal position and oblique sagittal position. At last, manual labeling of images by radiologists is considered the most accurate method for training models, but this is a labor-intensive method and limits the number of MRI cervical spine studies available for training.

In conclusion, we demonstrated that our DL model is reliable and may be used to quickly assess CSS at MRI. In clinical practice, the diagnosis of CSS still relies on the subjective opinion of the reporting radiologist. Our DL model showed comparable performance with subspecialist radiologists for detection and classification of the central canal and neural foraminal stenosis at cervical spine MRI with significant time savings. This method can provide a reference diagnosis for cervical spinal stenosis and also be assessed for the longitudinal follow-up of CSS at MR.

Abbreviations

CSS

Cervical spinal stenosis

SCS

Spinal canal stenosis

NFS

Neural foramen stenosis

DL

Deep learning

CNN

Convolutional neural network

ROIs

Region of interests
ACC
Accuracy
AUC
Area under curve
IoU
Intersection over Union
L
Left neural foramen
R
Right neural foramen
C
Central canal

Declarations

Ethics approval and consent to participate The methods were performed in accordance with relevant guidelines and regulations and approved by Peking University Third Hospital Medical Science Research Ethic Committee.

Consent for publication Patient consent was waived due to retrospective study.

Availability of data and materials The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Competing interests The authors declare no conflict of interest.

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Authors' contributions All authors contributed extensively to the work presented in this paper. Their roles are given below.

Design of the work: NL, SQJ, HSY.

Acquisition of data: QZW, YYC, KL, WLZ, XYX, YZ, HQO, LJ.

Interpretation of data: MYY, FYM, GWC, XHS.

Drafting/Critical revision: ELZ, MYY, YL, NL, XHS.

Final approval: NL, HSY.

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References

1. Lebl D R, Bono C M. Update on the Diagnosis and Management of Cervical Spondylotic Myelopathy. *J Am Acad Orthop Surg*. 2015;23(11):648-660.
2. Martin A R, Tadokoro N, Tetreault L, et al. Imaging Evaluation of Degenerative Cervical Myelopathy: Current State of the Art and Future Directions. *Neurosurg Clin N Am*. 2018;29(1):33-45.
3. Tracy J A, Bartleson J D. Cervical spondylotic myelopathy. *Neurologist*. 2010;16(3):176-187.
4. Behrbalk E, Salame K, Regev G J, et al. Delayed diagnosis of cervical spondylotic myelopathy by primary care physicians. *Neurosurg Focus*. 2013;35(1):E1.
5. Edwards C C, 2nd, Riew K D, Anderson P A, Hilibrand A S, Vaccaro A F. Cervical myelopathy. current diagnostic and treatment strategies. *Spine J*. 2003;3(1):68-81.
6. Tetreault L A, Cote P, Kopjar B, et al. A clinical prediction model to assess surgical outcome in patients with cervical spondylotic myelopathy: internal and external validations using the prospective multicenter AOSpine North American and international datasets of 743 patients. *Spine J*. 2015;15(3):388-397.
7. Michopoulou S K, Costaridou L, Panagiotopoulos E, et al. Atlas-based segmentation of degenerated lumbar intervertebral discs from MR images of the spine. *IEEE Trans Biomed Eng*. 2009;56(9):2225-2231.
8. Berg-Johansen B, Han M, Fields A J, et al. Cartilage Endplate Thickness Variation Measured by Ultrashort Echo-Time MRI Is Associated With Adjacent Disc Degeneration. *Spine (Phila Pa 1976)*. 2018;43(10):E592-E600.
9. Zhou Y, Liu Y, Chen Q, Gu G, Sui X. Automatic Lumbar MRI Detection and Identification Based on Deep Learning. *J Digit Imaging*. 2019;32(3):513-520.
10. Jamaludin A, Lootus M, Kadir T, et al. ISSLS PRIZE IN BIOENGINEERING SCIENCE 2017: Automation of reading of radiological features from magnetic resonance images (MRIs) of the lumbar spine without human intervention is comparable with an expert radiologist. *Eur Spine J*. 2017;26(5):1374-1383.
11. Jamaludin A, Kadir T, Zisserman A. SpineNet: Automated classification and evidence visualization in spinal MRIs. *Med Image Anal*. 2017;41:63-73.
12. Lu J-T, Pedemonte S, Bizzo B, et al. DeepSPINE: Automated Lumbar Vertebral Segmentation, Disc-level Designation, and Spinal Stenosis Grading Using Deep Learning 2018 July 01, 2018:

- [arXiv:1807.10215 p.]. Available from: <https://ui.adsabs.harvard.edu/abs/2018arXiv180710215L>.
13. Hallinan J, Zhu L, Yang K, et al. Deep Learning Model for Automated Detection and Classification of Central Canal, Lateral Recess, and Neural Foraminal Stenosis at Lumbar Spine MRI. *Radiology*. 2021;300(1):130-138.
 14. Hayashi D. Deep Learning for Lumbar Spine MRI Reporting: A Welcome Tool for Radiologists. *Radiology*. 2021;300(1):139-140.
 15. Merali Z, Wang J Z, Badhiwala J H, et al. A deep learning model for detection of cervical spinal cord compression in MRI scans. *Sci Rep*. 2021;11(1):10473.
 16. Ma S, Huang Y, Che X, Gu R. Faster RCNN-based detection of cervical spinal cord injury and disc degeneration. *J Appl Clin Med Phys*. 2020;21(9):235-243.
 17. Litjens G, Kooi T, Bejnordi B E, et al. A survey on deep learning in medical image analysis. *Med Image Anal*. 2017;42:60-88.
 18. Ren S, He K, Girshick R, Sun J. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *IEEE Trans Pattern Anal Mach Intell*. 2017;39(6):1137-1149.
 19. Kang Y, Lee J W, Koh Y H, et al. New MRI grading system for the cervical canal stenosis. *AJR Am J Roentgenol*. 2011;197(1):W134-140.
 20. Park H J, Kim S S, Lee S Y, et al. A practical MRI grading system for cervical foraminal stenosis based on oblique sagittal images. *Br J Radiol*. 2013;86(1025):20120515.
 21. Kushchayev S V, Glushko T, Jarraya M, et al. ABCs of the degenerative spine. *Insights Imaging*. 2018;9(2):253-274.
 22. Hutchins J, Hebelka H, Lagerstrand K, et al. A systematic review of validated classification systems for cervical and lumbar spinal foraminal stenosis based on magnetic resonance imaging. *Eur Spine J*. 2022;31(6):1358-1369.
 23. Sampath P, Bendebba M, Davis J D, Ducker T B. Outcome of patients treated for cervical myelopathy. A prospective, multicenter study with independent clinical review. *Spine (Phila Pa 1976)*. 2000;25(6):670-676.
 24. Meyer F, Borm W, Thome C. Degenerative cervical spinal stenosis: current strategies in diagnosis and treatment. *Dtsch Arztebl Int*. 2008;105(20):366-372.
 25. Kadanka Z, Mares M, Bednanik J, et al. Approaches to spondylotic cervical myelopathy: conservative versus surgical results in a 3-year follow-up study. *Spine (Phila Pa 1976)*. 2002;27(20):2205-2210; discussion 2210-2201.
 26. Takahashi M, Sakamoto Y, Miyawaki M, Bussaka H. Increased MR signal intensity secondary to chronic cervical cord compression. *Neuroradiology*. 1987;29(6):550-556.
 27. Sun B, Xu C, Zhang Y, et al. Intervertebral Foramen Width Is an Important Factor in Deciding Additional Uncinate Process Resection in ACDF-a Retrospective Study. *Front Surg*. 2021;8:626344.
 28. He K, Sun J. Convolutional Neural Networks at Constrained Time Cost. 2014.

29. Kim W, Ahn K S, Kang C H, Kang W Y, Yang K S. Comparison of MRI grading for cervical neural foraminal stenosis based on axial and oblique sagittal images: Concordance and reliability study. Clin Imaging. 2017;43:165-169.

Figures

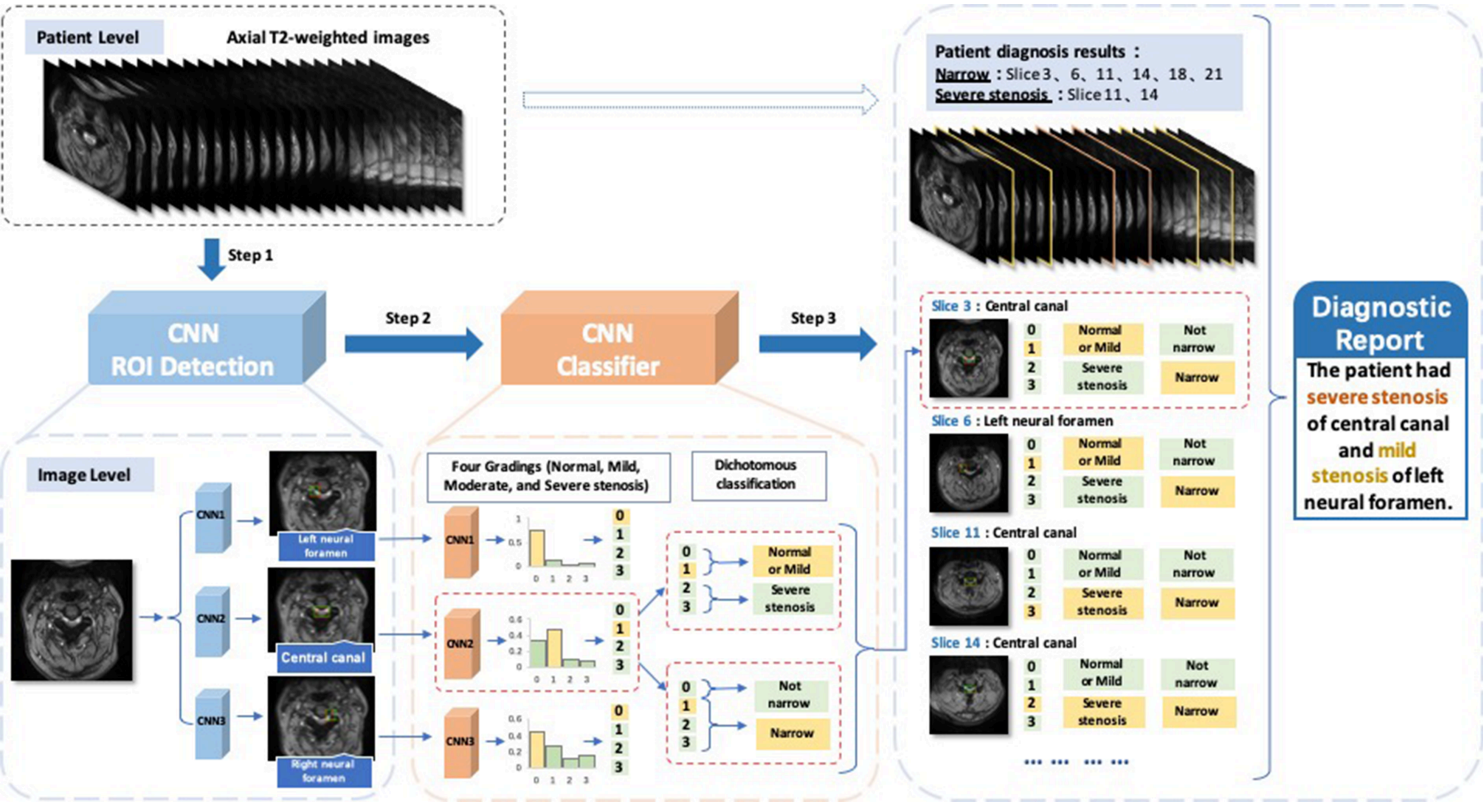


Figure 1

The flow charts of DL models for automated detection and classification of central canal and neural foraminal stenosis at cervical spine MRI.

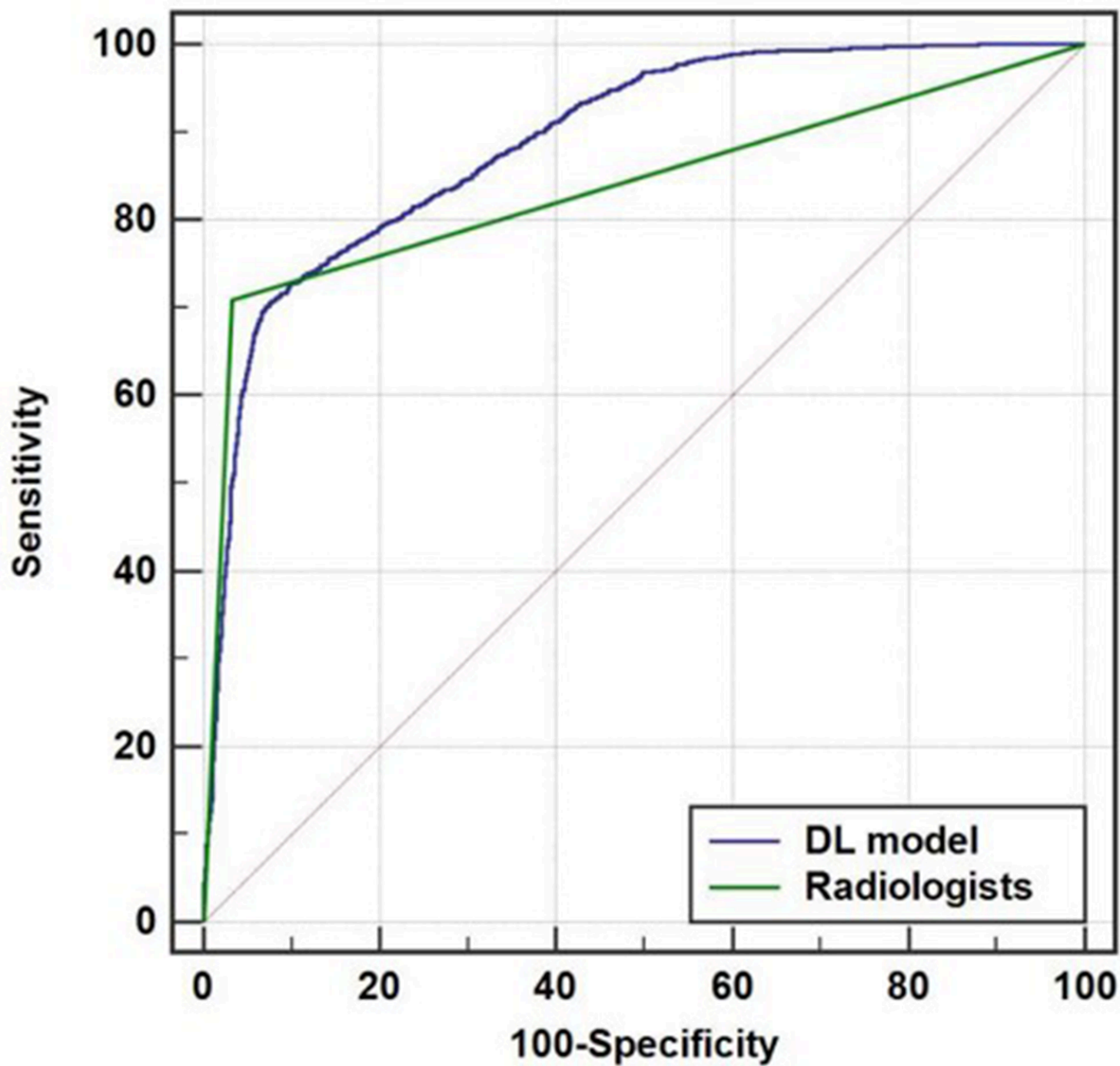


Figure 2

The ROC curve on dichotomous classification of the neural foramen (normal or mild vs moderate or severe).

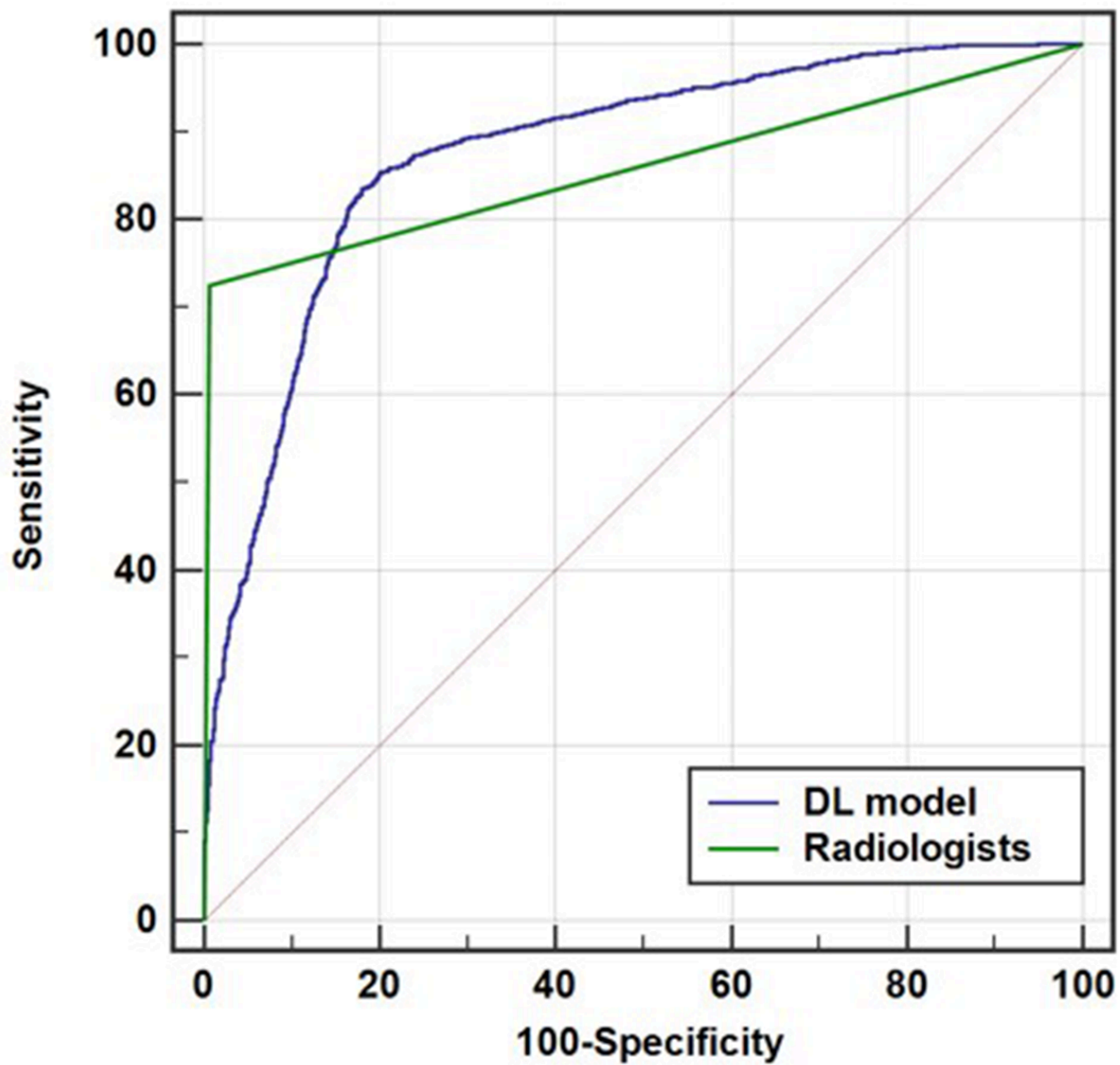


Figure 3

The ROC curve on dichotomous classification of central canal (normal or mild vs moderate or severe).